

Technology

“The Core Issue With ADAS Is Not Just Automotive Technology But An AI Challenge”

India's roads push ADAS to its limits. Can we innovate solutions and lead in safer driving? In this Q&A, EFY's Mukul Kumar explores this with Gagandeep Reehal, the 23-year-old CEO and Co-founder of Minus Zero.

GAGANDEEP REEHAL
CEO AND CO-FOUNDER, MINUS ZERO



What are the different categories of ADAS relevant to India, and why has its adoption been challenging compared to other countries?

Advanced driver assistance systems (ADAS) vary in complexity from level 0 to level 5 autonomy. Level 0 (L0) represents standard vehicles where the driver controls all functions. Level 1 (L1) includes basic warning features, such as lane departure alerts. Level 2 (L2) introduces automation, including lane-keeping assist and adaptive cruise control, using systems like Mobileye, now prevalent in vehicles from brands such as Mahindra.

The challenge of adopting ADAS in India, compared to other countries, arises from the country's unique road conditions. Inconsistent or absent lane markings, diverse vehicle types, and unpredictable road scenarios hinder the effectiveness of ADAS technologies, which rely on consistently detecting markers and obstacles. Moreover, substantial global investments in ADAS research and development (R&D) have not yet translated effectively to

Indian roads, where the variability and complexity far exceed typical test environments. This situation has posed significant challenges in creating reliable systems that can adapt to India's diverse traffic conditions, like the leap from basic chatbots to more advanced AI models like ChatGPT and large language models (LLMs).

Can you describe your journey in identifying and addressing the challenges with ADAS, especially in environments without clear infrastructure, such as missing lane markings in India?

Absolutely. The core issue with ADAS is not just automotive technology but an AI challenge—how to build a model that can understand the world and make decisions like a human mind. At its core, AI involves interpreting raw data and recognising patterns on a large scale. Traditional AI struggles with unexpected scenarios because it lacks human-like reasoning. For example, an AI system might misinterpret a red sunset as a traffic light due to similar visual cues, leading

to potential accidents. This limitation became apparent in incidents where AI-driven vehicles mistakenly braked for sunsets, causing accidents.

Recognising this, it became clear that automotive firms alone could not solve these problems. The breakthrough was developing a foundational model for navigation (like foundational language models developed by companies like OpenAI) that could ‘think’ more like humans, recognising objects in diverse and unpredictable conditions. This approach became our core intellectual property, bridging rigorous research with practical application in the industry.

How do electronics integrate into your operations, particularly given your startup's resemblance to an MSME? What advantages have allowed you, unlike major automobile companies, to overcome challenges effectively?

The fundamental distinction lies in the differences between hardware and software development processes. Automobile manufacturers may